

Software Requirements Specification (SRS)

Smart Complaint Management System

Field	Detail
Project	Smart Complaint Management System (SCMS)
Team	Team 8
Project Description	A platform for lodging and tracking complaints with admin resolution.
Implementation Language	Python 3.11 (Django 5.x / Django REST Framework)
Version	1.0
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Revision History

Version	Date	Author	Change summary
0.1	30-08-2026	Team 8	Initial draft — scope, actors and high-level feature list
0.2	03-09-2026	Team 8	Functional and non-functional requirements added with acceptance criteria
1.0	06-09-2026	Team 8	Security section, use-case diagrams and RTM added; released for review

Team 8 — Roles and Responsibilities

Project: Smart Complaint Management System (SCMS)

Name	Role	SRN	Responsibility
ARYAMAN D K	Product Owner / Team Lead	PES1UG24AM344	Gather and define requirements, author the SRS, prioritise features and maintain the Jira backlog
VIKAS GOWDA L	Backend Developer	PES1UG24AM329	Build Django models, REST APIs, complaint routing and SLA escalation logic
V S R VINAY VARMA	Frontend Developer	PES1UG24AM317	Build responsive UI templates, complaint forms, status tracking and dashboards
Shridhar Umesh Golipalle	QA Lead / DevOps	PES1UG25AM814	Prepare the test plan, design and execute test cases, log defects, handle database and deployment

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1. Introduction

1.1 Purpose

This document is the Software Requirements Specification for the Smart Complaint Management System (SCMS), a web-based platform on which citizens lodge complaints and administrators track and resolve them. It states the functional requirements, non-functional requirements, external interfaces, security requirements and verification criteria that the delivered system must satisfy. It is the agreed reference against which the design, implementation and acceptance testing of the mini project will be judged.

1.2 Scope

SCMS covers the complete life cycle of a complaint: registration and authentication of users, lodging of a complaint with supporting evidence, automatic routing to the responsible department, assignment to a support agent, SLA-based escalation, status tracking by the complainant, resolution with proof, reopening, closure with feedback, and administrative reporting and audit.

The following are outside the scope of this release:

- Payment or fee collection of any kind.
- Native Android or iOS applications; the system is delivered as a responsive web application.
- Integration with external government or municipal grievance systems.
- The internal implementation of the e-mail and SMS gateways, which are consumed as third-party services.

1.3 Audience

Developers of Team 8, QA engineers, the project guide and course coordinator, evaluators assessing the mini project, and any team that maintains the system after handover.

1.4 Definitions, acronyms and abbreviations

Term	Meaning
SCMS	Smart Complaint Management System — the product described by this document
Complainant	A citizen who lodges a complaint, whether registered or guest
Support Agent	A departmental staff member to whom complaints are assigned for resolution
Department Admin	The user who assigns work within a department and monitors its SLA performance
System Admin	The user who manages master data, users, roles and system-wide reports
SLA	Service Level Agreement — the maximum permitted time to resolve a complaint of a given category
FR / NFR	Functional Requirement / Non-Functional Requirement
RTM	Requirements Traceability Matrix
RBAC	Role-Based Access Control
OTP	One-Time Password sent to a mobile number for verification
ORM	Object-Relational Mapper — the Django database abstraction layer
API	Application Programming Interface
TLS	Transport Layer Security
CSRF / XSS / SQLi	Cross-Site Request Forgery / Cross-Site Scripting / SQL Injection
WCAG	Web Content Accessibility Guidelines

Term	Meaning
DPDP Act	Digital Personal Data Protection Act, 2023 (India)

2. Overall description

2.1 Product perspective

SCMS is a new, self-contained web application rather than a replacement for an existing product. It follows a three-tier architecture: a responsive browser-based presentation tier, a Django application tier holding the business rules, and a PostgreSQL data tier. A background worker handles scheduled and asynchronous work such as SLA escalation and notification dispatch. The application depends on three external services — an SMTP e-mail service, an SMS gateway and an object store for evidence files.

2.2 Major product functions

- Register, verify and authenticate users across four roles.
- Lodge a complaint with category, description, location and evidence attachments.
- Issue a unique complaint reference ID and acknowledge it to the complainant.
- Automatically route the complaint to the responsible department and assign a priority.
- Assign and reassign complaints to support agents within a department.
- Escalate complaints automatically when the category SLA is breached.
- Track complaint status and exchange comments between complainant and agent.
- Record resolution remarks and proof, and allow the complainant to reopen or rate the outcome.
- Send e-mail and SMS notifications at each significant event.
- Provide role-scoped dashboards, search, filtering and CSV/PDF export.
- Maintain an append-only audit trail of every state change.

2.3 User roles and characteristics

Role	Characteristics	Primary expectations
Complainant (Citizen / Guest)	General public, varying digital literacy, often on a mobile browser	A short lodging form, a reference ID that works, and honest status updates
Support Agent	Departmental staff, works through a daily queue, moderate system familiarity	A clear queue, complete complaint context, and few clicks to update status
Department Admin	Supervises a department, accountable for SLA compliance	Workload visibility, easy reassignment, and early warning of SLA breaches
System Admin	Technically competent administrator of the whole platform	Control of master data, users and roles, plus system-wide reports and audit access

2.4 Operating environment

Layer	Environment
Client	Chrome, Firefox, Edge or Safari (latest two versions) on desktop or mobile; screen width 320 px and above
Application server	Linux (Ubuntu 22.04 LTS), Python 3.11+, Django 5.x with Django REST Framework, served by Gunicorn behind Nginx
Database	PostgreSQL 14 or higher
Background processing	Celery workers with Redis as broker; Celery Beat for scheduled SLA and clean-up jobs
Storage	Local or S3-compatible object store for evidence files, located outside the web root
External services	SMTP relay for e-mail; HTTP-based SMS gateway for OTP and alerts

2.5 Constraints

- The implementation language is fixed to Python, using the Django framework, as assigned to Team 8.
- Handling of complainant personal data must be consistent with the Digital Personal Data Protection Act, 2023.
- The system must run within the modest compute budget of a student deployment: 2 vCPU and 4 GB RAM per application instance.
- Only open-source libraries with permissive licences may be used.
- The project must be delivered within the academic mini-project schedule, with the SRS baselined on 06-09-2025.
- All source code and documentation must be maintained in the team's GitHub mini-project repository.

3. External interface requirements

3.1 User interfaces

The system presents a single responsive web interface whose navigation adapts to the signed-in role. The following screens are required:

- Public: home page, registration, login, guest complaint form, and track-by-reference-ID page.
- Complainant portal: lodge complaint, my complaints list, complaint detail with status timeline and comments, reopen and rating screens.
- Agent portal: assigned queue with filters, complaint detail, status update and resolution screen.
- Department Admin portal: department queue, assignment and reassignment screen, SLA breach view, department dashboard.
- System Admin portal: master data management, user and role management, system dashboard, audit trail viewer and export screen.

Every screen must show validation messages next to the field concerned, must be operable by keyboard alone, and must meet the contrast requirements stated in SCMS-NF-004.

3.2 Hardware interfaces

SCMS requires no special-purpose hardware. It interacts only with standard client devices — desktop, laptop, tablet or smartphone — through the browser, and with the camera or file system of the client device when the complainant attaches evidence, using the standard HTML file input. The server environment requires a minimum of 2 vCPU, 4 GB RAM and 40 GB of storage per application instance.

3.3 Software interfaces

Interface	Type / Protocol	Purpose and data exchanged
PostgreSQL 14+	TCP, via Django ORM	Persistent storage of users, complaints, comments, attachments metadata, master data and the audit trail
SMTP e-mail service	SMTP over TLS, port 587	Delivery of acknowledgement, status change, escalation and password reset e-mails
SMS gateway	REST / JSON over HTTPS	Delivery of registration OTP and status-change alerts; sends mobile number and message text, receives a delivery status
Object store (local or S3-compatible)	Filesystem API or HTTPS	Storage and retrieval of evidence and resolution-proof files
Redis	TCP	Message broker for Celery tasks and cache for rate-limit counters
SCMS REST API	REST / JSON over HTTPS	Internal interface between the browser client and the application tier; also the extension point for any future mobile client

3.4 Communications interfaces

- All browser-to-server traffic uses HTTPS with TLS 1.2 or higher; plain HTTP requests are redirected to HTTPS.
- Application-to-gateway calls (SMS, e-mail) use TLS and authenticate with an API key or credentials held in server-side environment variables, never in source code.
- API requests and responses use JSON encoded in UTF-8, so that regional-language complaint text is preserved.
- Outbound gateway calls use a 10-second timeout with an exponential-backoff retry policy, as required by SCMS-F-022.
- Session state is carried in a signed cookie marked Secure, HttpOnly and SameSite=Lax.

4. System features (detailed)

Each requirement below carries an acceptance criterion and a reference test case. Functional requirement IDs follow the pattern SCMS-F-####. This section defines 28 functional requirements across six feature groups.

4.1 User Registration, Authentication and Access Control

Description: establishes who may use the system and what each role is permitted to do. It is a prerequisite for every other feature group.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SCMS-F-001	The system shall allow a citizen to register using full name, e-mail ID and mobile number, and shall verify the mobile number through a 6-digit OTP valid for 10 minutes.	Functional	High	Citizen / Product Owner	AC-001: Account is created only after correct OTP entry; expired or wrong OTP is rejected. Test: TC-Auth-01	Requires SMS gateway
SCMS-F-002	The system shall authenticate registered users using e-mail ID and password, and shall terminate the session after 30 minutes of inactivity.	Functional	High	Security	AC-002: Valid credentials create an authenticated session; session expires after 30 min idle. Test: TC-Auth-02	Depends on SCMS-F-001
SCMS-F-003	The system shall enforce role-based access control for four roles — Complainant, Support Agent, Department Admin and System Admin.	Functional	High	Security / Admin	AC-003: A user accessing a page outside their role receives HTTP 403. Test: TC-Auth-03	Basis for SCMS-SR-003
SCMS-F-004	The system shall allow password reset through a single-use link e-mailed to the registered address and valid for 15 minutes.	Functional	Medium	Citizen	AC-004: Link works once; reuse or use after 15 min is refused. Test: TC-Auth-04	Requires SMTP service
SCMS-F-005	The system shall lock a user account for 15 minutes after 5 consecutive failed login attempts and shall record an audit event.	Functional	High	Security	AC-005: 6th attempt is blocked with a message; audit entry created. Test: TC-Auth-05	Links to SCMS-F-027

4.2 Complaint Lodging and Evidence Capture

Description: allows a registered or guest complainant to submit a complaint with supporting evidence and receive a reference ID for later tracking.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SCMS-F-006	The system shall allow an authenticated complainant to lodge a complaint capturing category, sub-category, subject, description (max 2000 characters) and location.	Functional	High	Citizen / Business	AC-006: Complaint is saved only when all mandatory fields are valid; validation errors are shown field-wise. Test: TC-Lodge-01	Core feature
SCMS-F-007	The system shall generate a unique complaint reference ID in the format SCMS-YYYY-NNNNNN on successful submission and display it to the complainant.	Functional	High	Business	AC-007: Every complaint has a distinct reference ID; no duplicates across 10,000 submissions. Test: TC-Lodge-02	DB unique constraint

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SCMS-F-008	The system shall accept up to 3 evidence attachments per complaint in JPG, PNG or PDF format, each not exceeding 5 MB.	Functional	Medium	Citizen / Operations	AC-008: A 4th file or a 6 MB file or a .exe file is rejected with a clear message. Test: TC-Lodge-03	See SCMS-SR-005
SCMS-F-009	The system shall allow a guest (unregistered) user to lodge a complaint after mobile OTP verification and shall issue a tracking token for later access.	Functional	Medium	Citizen	AC-009: Guest complaint is created and the token retrieves it without login. Test: TC-Lodge-04	Needs CAPTCHA (SCMS-SR-007)
SCMS-F-010	The system shall let a complainant save a complaint as a draft and resume it within 7 days, after which the draft is purged.	Functional	Low	Citizen	AC-010: Draft is retrievable before 7 days and absent afterwards. Test: TC-Lodge-05	Scheduled clean-up job

4.3 Routing, Assignment, Prioritisation and Escalation

Description: moves a complaint from submission to an accountable owner, sets its urgency, and escalates it when the SLA is at risk.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SCMS-F-011	The system shall automatically route a newly lodged complaint to the responsible department using the configured category-to-department mapping.	Functional	High	Operations	AC-011: Complaint appears in the mapped department queue within 5 seconds of submission. Test: TC-Route-01	Depends on SCMS-F-023
SCMS-F-012	The system shall allow a Department Admin to assign or reassign a complaint to a Support Agent, with a mandatory reason recorded for every reassignment.	Functional	High	Department Admin	AC-012: Reassignment without a reason is blocked; reason appears in the audit trail. Test: TC-Route-02	Links to SCMS-F-027
SCMS-F-013	The system shall assign a default priority (Low, Medium, High, Critical) from the category priority matrix, which a Department Admin may override with a recorded reason.	Functional	Medium	Operations	AC-013: Default priority matches the matrix; override is logged with the old and new value. Test: TC-Route-03	Configurable master data
SCMS-F-014	The system shall automatically escalate a complaint to the Department Admin when its category SLA is breached, and to the System Admin at twice the SLA duration.	Functional	High	Business / Operations	AC-014: Escalation is raised within 15 minutes of breach and both levels are notified. Test: TC-Route-04	Requires Celery Beat scheduler
SCMS-F-015	The system shall flag a complaint as a potential duplicate when the same complainant files in the same category with at least 70% description similarity within 30 days.	Functional	Low	Operations	AC-015: Duplicate flag is shown to the agent, who may confirm or dismiss it. Test: TC-Route-05	Similarity via trigram index

4.4 Tracking, Resolution and Feedback

Description: keeps the complainant informed, governs how a complaint is resolved and closed, and captures satisfaction with the outcome.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SCMS-F-016	The system shall allow a complaint to be tracked by reference ID through the states Open, Assigned, In Progress, Pending Information, Resolved, Closed and Reopened.	Functional	High	Citizen	AC-016: Status page shows the current state and a dated history of transitions. Test: TC-Track-01	State machine validation
SCMS-F-017	The system shall provide threaded comments between the complainant and the assigned Support Agent on each complaint.	Functional	Medium	Citizen / Agent	AC-017: Comment is visible to both parties and to no other complainant. Test: TC-Track-02	Object-level check (SCMS-SR-003)
SCMS-F-018	The system shall require a Support Agent to record resolution remarks of at least 50 characters and attach at least one proof file before moving a complaint to Resolved.	Functional	High	Quality / Operations	AC-018: Transition to Resolved fails without remarks and proof. Test: TC-Res-01	Depends on SCMS-F-008
SCMS-F-019	The system shall allow a complainant to reopen a resolved complaint once within 7 days, after which the complaint is closed automatically.	Functional	Medium	Citizen	AC-019: Reopen works on day 6, is refused on day 8; auto-closure runs on day 8. Test: TC-Res-02	Scheduled job
SCMS-F-020	The system shall capture a 1-to-5 star satisfaction rating and an optional comment from the complainant when a complaint is closed.	Functional	Low	Business	AC-020: Rating is stored and reflected in the department dashboard average. Test: TC-Res-03	Feeds SCMS-F-025

4.5 Notification

Description: informs complainants and staff of significant events over e-mail and SMS, reliably and without silent failure.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SCMS-F-021	The system shall send e-mail and SMS notifications on lodging, assignment, status change, resolution and escalation within 2 minutes of the event.	Functional	High	Citizen / Operations	AC-021: Notification is dispatched within 2 min for each of the five events. Test: TC-Notif-01	SMTP and SMS gateway
SCMS-F-022	The system shall retry a failed notification up to 3 times with exponential backoff and shall log a permanent failure for administrator review.	Functional	Medium	Operations	AC-022: With the gateway stubbed to fail, 3 retries occur and a failure record is written. Test: TC-Notif-02	Celery retry policy

4.6 Administration, Reporting and Audit

Description: gives administrators control of configuration and users, visibility over performance, and a tamper-evident record of all activity.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SCMS-F-023	The system shall allow the System Admin to create, edit and deactivate master data — departments, categories, category-to-department mapping, SLA durations and the priority matrix.	Functional	High	System Admin	AC-023: Master data changes take effect on newly lodged complaints without a redeploy. Test: TC-Admin-01	Prerequisite for SCMS-F-011
SCMS-F-024	The system shall allow the System Admin to create users, change their roles and deactivate accounts; a deactivated user shall not be able to log in.	Functional	High	System Admin	AC-024: Deactivated user login is refused; open complaints are reassignable. Test: TC-Admin-02	Links to SCMS-F-003
SCMS-F-025	The system shall provide a role-scoped dashboard showing complaint counts by status, department, priority, SLA breach and average satisfaction rating.	Functional	Medium	Department Admin / Business	AC-025: Agent sees only own queue; Department Admin sees own department; System Admin sees all. Test: TC-Rep-01	Depends on SCMS-F-020
SCMS-F-026	The system shall allow an administrator to export the filtered complaint list to CSV and PDF.	Functional	Low	Business	AC-026: Exported file row count and column values match the on-screen filtered list. Test: TC-Rep-02	Uses ReportLab
SCMS-F-027	The system shall maintain an append-only audit trail recording actor, action, timestamp, source IP and before/after values for every complaint state change.	Functional	High	Audit / Security	AC-027: Audit row exists for each transition and cannot be edited or deleted through the application. Test: TC-Audit-01	Supports SCMS-SR-006
SCMS-F-028	The system shall allow searching and filtering of complaints by reference ID, status, category, department, date range and free-text keyword.	Functional	Medium	Agent / Admin	AC-028: Each filter and combination returns the correct result set within 2 seconds. Test: TC-Rep-03	DB indexes required

5. Non-functional requirements (detailed)

The requirements below are measurable and each is tied to a test in the test plan. IDs follow the pattern SCMS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
SCMS-NF-001	The 95th percentile server response time for all pages and API calls shall not exceed 2 seconds with 200 concurrent users.	Performance	High	JMeter run of 200 virtual users shows p95 ≤ 2 s. Test: TC-Perf-01
SCMS-NF-002	The system shall be available 99.5% of the time each calendar month, excluding announced maintenance windows.	Reliability / Availability	High	Uptime monitor report shows ≥ 99.5% monthly. Test: TC-Ops-01
SCMS-NF-003	The system shall support 50,000 complaints per year and 500 concurrent sessions, and shall scale horizontally behind a load balancer without code change.	Scalability	Medium	Load test at 500 sessions passes with two application instances. Test: TC-Perf-02
SCMS-NF-004	The web interface shall conform to WCAG 2.1 Level AA, render correctly from 320 px width upwards, and support English plus one regional language.	Usability / Accessibility	Medium	Axe accessibility scan reports no Level AA violations. Test: TC-UX-01
SCMS-NF-005	The codebase shall follow PEP 8, be organised into independent modules, and maintain at least 70% line coverage from automated unit tests.	Maintainability	Medium	flake8 reports no errors and coverage report shows ≥ 70%. Test: TC-Maint-01
SCMS-NF-006	The system shall run on any Linux host with Python 3.11+ and PostgreSQL 14+, and shall be deployable as Docker containers.	Portability	Low	A clean container build starts and passes the smoke test suite. Test: TC-Port-01
SCMS-NF-007	Complaint records and audit logs shall be retained for 3 years, then archived and purged by a scheduled job.	Auditability / Data retention	Medium	Records older than 3 years are archived and removed from the live table. Test: TC-Audit-02

5.1 Security

5.1.1 Security objectives

Four security objectives govern the design of SCMS. Every security requirement in section 5.1.2 traces to at least one of them.

ID	Objective	Rationale
SO-01	Confidentiality of complainant data	Complaints contain personal identifiers, location details and photographic evidence. This data must be readable only by the complainant and by the staff handling that specific complaint, both in transit and at rest.
SO-02	Integrity and non-repudiation of the complaint life cycle	A complaint record, its status history and its evidence must not be altered or deleted without an attributable, permanent record, so that resolution claims can be relied upon during review.
SO-03	Accountability through least-privilege access	Each of the four roles must hold only the permissions its work requires, and every privileged action must be traceable to an identified user.
SO-04	Availability and resistance to abuse	A public complaint form is an open target for spam and flooding. The system must remain available to genuine users under such load.

5.1.2 Security requirements

IDs follow the pattern SCMS-SR-###. The traceability of each requirement to a security objective is shown in the RTM in section 8.

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
SCMS-SR-001	All client-server and server-gateway connections shall use TLS 1.2 or higher; HTTP requests shall be redirected to HTTPS and HSTS shall be enabled.	Security	High	SSL scan reports TLS 1.2+ only and HSTS present. Test: TC-Sec-01
SCMS-SR-002	User passwords shall be stored using Argon2id (or bcrypt with cost factor ≥ 12) and shall never be written to logs, displayed or transmitted in plain text.	Security	High	Database inspection shows only hashes; log grep for password returns nothing. Test: TC-Sec-02
SCMS-SR-003	Access shall be deny-by-default; every complaint-level operation shall additionally verify that the requesting user owns or is assigned to that complaint.	Security	High	Changing the complaint ID in the URL to another user's complaint returns 403. Test: TC-Sec-03
SCMS-SR-004	All input shall be validated server-side, database access shall use parameterised ORM queries, output shall be escaped, and state-changing forms shall carry CSRF tokens.	Security	High	OWASP ZAP scan reports no High or Medium SQLi, XSS or CSRF findings. Test: TC-Sec-04
SCMS-SR-005	Uploaded evidence files shall be stored outside the web root with randomised filenames, validated by both MIME type and extension, scanned for malware, and served with Content-Disposition: attachment.	Security	High	An uploaded .php file is rejected; a stored file cannot be executed via a direct URL. Test: TC-Sec-05
SCMS-SR-006	The audit log shall be append-only; the application database account shall hold no UPDATE or DELETE privilege on the audit table.	Security	High	An attempted update of an audit row from the application account fails. Test: TC-Sec-06
SCMS-SR-007	The system shall rate-limit public endpoints to 5 complaint submissions per mobile number per hour and 10 login attempts per IP per 5 minutes, and shall require CAPTCHA on guest lodging.	Security	Medium	The 6th submission in an hour and the 11th login in 5 minutes are throttled. Test: TC-Sec-07

6. Quality attributes & Acceptance tests

6.1 Exit criteria for acceptance

- Every functional requirement of High priority is implemented and verified, with its acceptance criterion demonstrably met.
- No critical or high-severity defect remains open.
- No non-functional requirement fails its stated measurement.
- All seven security requirements pass their tests, and the OWASP ZAP baseline scan reports no High or Medium finding.
- The RTM in section 8 shows every requirement mapped to at least one test case, with status A (Accepted).
- Automated unit test line coverage is at least 70%.

6.2 Acceptance test suites

Suite	Requirements covered	Focus of testing
Authentication suite	SCMS-F-001 to SCMS-F-005	Registration, OTP, login, session expiry, role enforcement, reset and logout
Lodging suite	SCMS-F-006 to SCMS-F-010	Field validation, reference ID uniqueness, attachment limits, guest flow, drafts
Routing suite	SCMS-F-011 to SCMS-F-015	Category mapping, assignment, priority, SLA escalation timing, duplicate flagging
Resolution suite	SCMS-F-016 to SCMS-F-020	Status transitions, comments, resolution gating, reopen window, rating capture
Notification suite	SCMS-F-021, SCMS-F-022	Dispatch latency, channel coverage, retry and failure logging
Administration suite	SCMS-F-023 to SCMS-F-028	Master data effect, user lifecycle, dashboard scoping, export fidelity, audit completeness, search
Performance suite	SCMS-NF-001, SCMS-NF-003	Response time at 200 and 500 concurrent users, horizontal scaling behaviour
Accessibility suite	SCMS-NF-004	WCAG 2.1 AA conformance, keyboard navigation, responsive layout at 320 px
Security suite	SCMS-SR-001 to SCMS-SR-007	TLS configuration, password storage, authorisation bypass attempts, OWASP scan, file upload abuse, audit immutability, rate limiting

6.3 Quality attribute priorities

Where requirements conflict, the team resolves them in this order: security and data integrity first, then correctness of the complaint life cycle, then usability, then performance, and finally breadth of features. For example, if duplicate detection (SCMS-F-015, Low priority) degrades lodging response time beyond SCMS-NF-001, the detection is moved to a background task rather than relaxing the response-time target.

7. System models and diagrams

7.1 UML Use-Case diagrams

Two use-case diagrams are provided. The first covers the complainant-facing boundary of the system; the second covers staff and administrative use cases, including the two automated actors that act on the system without human intervention.

7.1.1 Use-Case Diagram 1 — Complainant-facing use cases

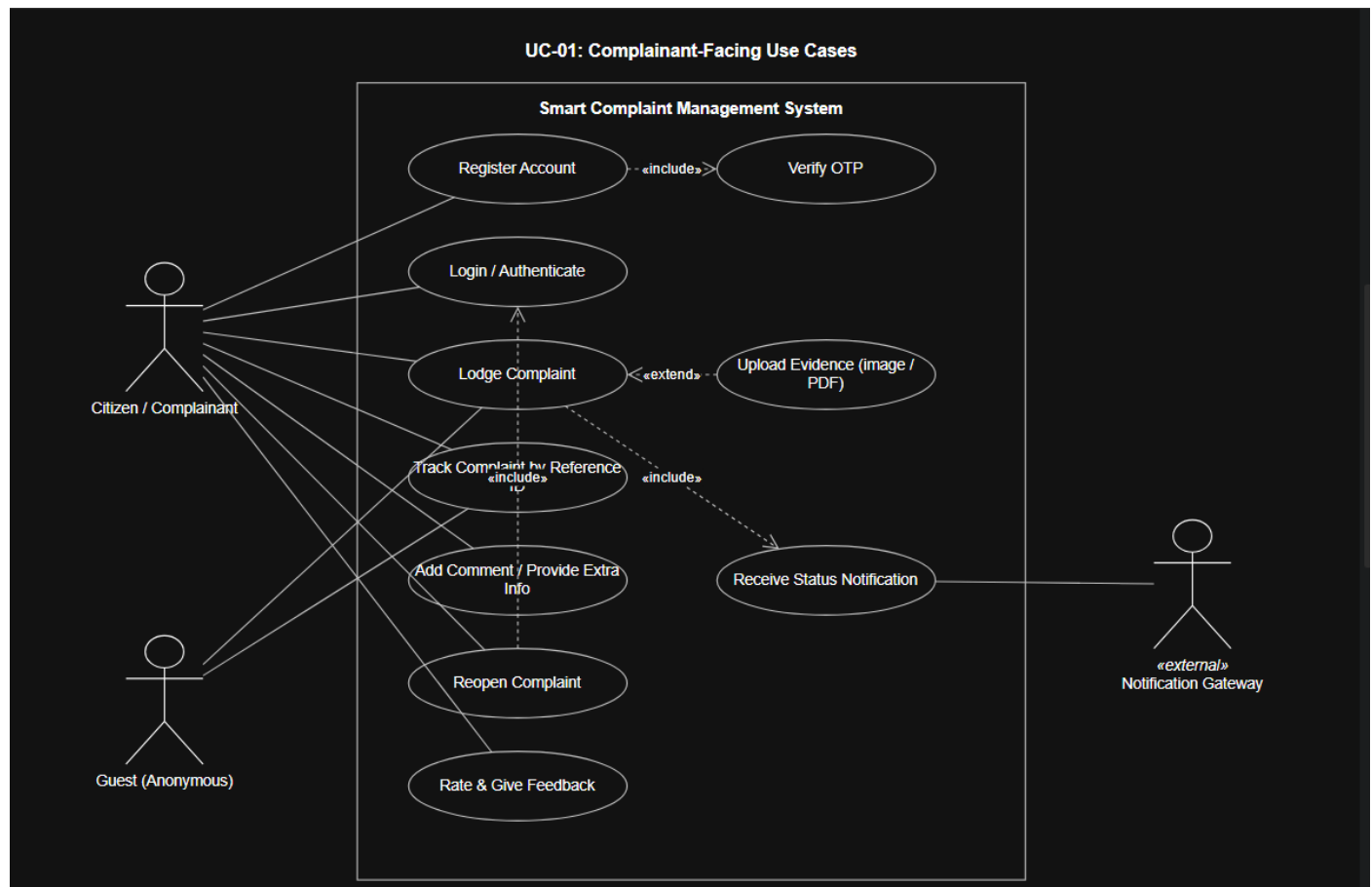


Figure 1: UC-01 — Complainant and guest interaction with SCMS. Registration includes OTP verification; lodging a complaint includes notification and may be extended by evidence upload.

Actors: Citizen/Complainant and Guest are human actors; the Notification Gateway is an external system actor that receives dispatch requests. The diagram shows that a guest may lodge and track a complaint but may not comment, reopen or rate, since those actions require an authenticated account.

7.1.2 Use-Case Diagram 2 — Staff and administrative use cases

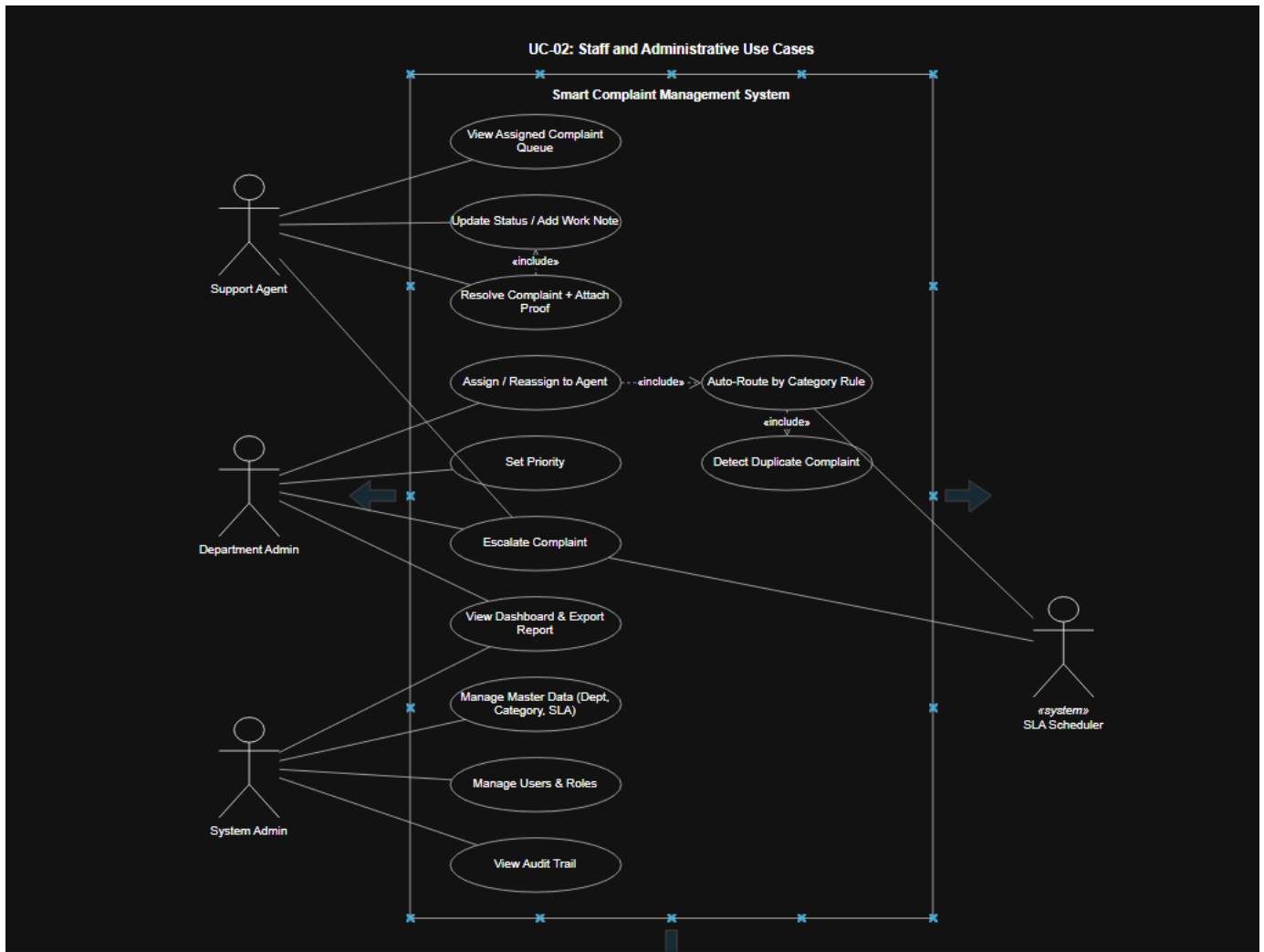


Figure 2: UC-02 — Support Agent, Department Admin, System Admin and the SLA Scheduler. Assignment includes auto-routing, which in turn includes duplicate detection; resolution includes a status update.

Actors: Support Agent, Department Admin and System Admin are human actors with progressively wider authority. The SLA Scheduler is a time-triggered system actor that initiates escalation (SCMS-F-014) and auto-routing (SCMS-F-011) without a user request, which is why it appears as an actor rather than as part of a human flow.

8. Requirements Traceability Matrix (RTM)

Status legend: N = Not started, P = In progress, A = Accepted. All requirements are at status N at the time this SRS is baselined.

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
SCMS-F-001	Register with OTP verification	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N	
SCMS-F-002	Login and session timeout	4.1 / DS-Auth-02	AuthModule	TC-Auth-02	N	
SCMS-F-003	Role-based access control	4.1 / DS-Auth-03	AuthModule	TC-Auth-03	N	Basis for SCMS-SR-003
SCMS-F-004	Password reset link	4.1 / DS-Auth-04	AuthModule	TC-Auth-04	N	
SCMS-F-005	Account lockout	4.1 / DS-Auth-05	AuthModule	TC-Auth-05	N	
SCMS-F-006	Lodge complaint	4.2 / DS-Comp-01	ComplaintModule	TC-Lodge-01	N	Core flow
SCMS-F-007	Generate reference ID	4.2 / DS-Comp-02	ComplaintModule	TC-Lodge-02	N	
SCMS-F-008	Evidence attachments	4.2 / DS-Comp-03	ComplaintModule	TC-Lodge-03	N	See SCMS-SR-005
SCMS-F-009	Guest complaint lodging	4.2 / DS-Comp-04	ComplaintModule	TC-Lodge-04	N	
SCMS-F-010	Draft save and resume	4.2 / DS-Comp-05	ComplaintModule	TC-Lodge-05	N	
SCMS-F-011	Auto-route to department	4.3 / DS-Route-01	RoutingModule	TC-Route-01	N	
SCMS-F-012	Assign / reassign to agent	4.3 / DS-Route-02	RoutingModule	TC-Route-02	N	
SCMS-F-013	Priority assignment	4.3 / DS-Route-03	RoutingModule	TC-Route-03	N	
SCMS-F-014	SLA escalation	4.3 / DS-Route-04	RoutingModule	TC-Route-04	N	Scheduler dependency
SCMS-F-015	Duplicate detection	4.3 / DS-Route-05	RoutingModule	TC-Route-05	N	
SCMS-F-016	Status tracking	4.4 / DS-Track-01	TrackingModule	TC-Track-01	N	
SCMS-F-017	Threaded comments	4.4 / DS-Track-02	TrackingModule	TC-Track-02	N	
SCMS-F-018	Resolution with proof	4.4 / DS-Track-03	TrackingModule	TC-Res-01	N	
SCMS-F-019	Reopen and auto-close	4.4 / DS-Track-04	TrackingModule	TC-Res-02	N	
SCMS-F-020	Satisfaction rating	4.4 / DS-Track-05	TrackingModule	TC-Res-03	N	

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
SCMS-F-021	Multi-channel notification	4.5 / DS-Notif-01	NotificationModule	TC-Notif-01	N	Gateway dependency
SCMS-F-022	Notification retry	4.5 / DS-Notif-02	NotificationModule	TC-Notif-02	N	
SCMS-F-023	Master data management	4.6 / DS-Admin-01	AdminModule	TC-Admin-01	N	
SCMS-F-024	User and role management	4.6 / DS-Admin-02	AdminModule	TC-Admin-02	N	
SCMS-F-025	Role-scoped dashboard	4.6 / DS-Rep-01	ReportModule	TC-Rep-01	N	
SCMS-F-026	CSV / PDF export	4.6 / DS-Rep-02	ReportModule	TC-Rep-02	N	
SCMS-F-027	Append-only audit trail	4.6 / DS-Audit-01	AuditModule	TC-Audit-01	N	Supports SCMS-SR-006
SCMS-F-028	Search and filter	4.6 / DS-Rep-03	ReportModule	TC-Rep-03	N	
SCMS-NF-001	Response time p95 ≤ 2 s	5 / DS-Perf-01	All modules	TC-Perf-01	N	
SCMS-NF-002	99.5% availability	5 / DS-Ops-01	Infrastructure	TC-Ops-01	N	
SCMS-NF-003	Scalability targets	5 / DS-Perf-02	Infrastructure	TC-Perf-02	N	
SCMS-NF-004	WCAG 2.1 AA and responsive UI	5 / DS-UX-01	WebUI	TC-UX-01	N	
SCMS-NF-005	Maintainability and coverage	5 / DS-Maint-01	All modules	TC-Maint-01	N	
SCMS-NF-006	Portability and containers	5 / DS-Port-01	Infrastructure	TC-Port-01	N	
SCMS-NF-007	3-year retention	5 / DS-Audit-02	AuditModule	TC-Audit-02	N	
SCMS-SR-001	TLS 1.2+ and HSTS	5.1.2 / DS-Sec-01	Infrastructure	TC-Sec-01	N	Objective SO-01
SCMS-SR-002	Password hashing	5.1.2 / DS-Sec-02	AuthModule	TC-Sec-02	N	Objective SO-01
SCMS-SR-003	Least-privilege authorisation	5.1.2 / DS-Sec-03	AuthModule	TC-Sec-03	N	Objective SO-03
SCMS-SR-004	Input validation and OWASP controls	5.1.2 / DS-Sec-04	WebUI / All modules	TC-Sec-04	N	Objective SO-02
SCMS-SR-005	Secure file handling	5.1.2 / DS-Sec-05	ComplaintModule	TC-Sec-05	N	Objective SO-01
SCMS-SR-006	Append-only audit log	5.1.2 / DS-Sec-06	AuditModule	TC-Sec-06	N	Objective SO-02
SCMS-SR-007	Rate limiting and CAPTCHA	5.1.2 / DS-Sec-07	WebUI / RoutingModule	TC-Sec-07	N	Objective SO-04

Coverage summary: 28 functional requirements, 7 non-functional requirements and 7 security requirements — 42 in total — each mapped to at least one test case and one implementation module.